



BTS-I-11.19-1455

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A**B.Tech. Degree I Semester Examination November 2019****CS/EC/IT 19-200-0105 B BASIC MECHANICAL ENGINEERING**
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Write a note on thermodynamic property. Why work is not considered as a property?
(b) "Carnot cycle is impractical and calculation of Carnot efficiency is insignificant". Analyze the statement and comment on it.
(c) With a neat sketch explain the working of a carburetor.
(d) Differentiate between heat pump and a refrigerator. Establish the connection between the CoP of refrigerator and heat pump.
(e) Define specific speed of a turbine. What is its significance?
(f) Explain the difference between positive displacement and rotodynamic pump. Give examples.
(g) What is heat treatment? What are the purposes of heat treatment?
(h) What are the classifications of gears? Explain.

PART B

(4 × 12 = 48)

- II. What do you mean by a thermodynamic process? Find the work done and internal energy change for an: (i) isochoric process (ii) isothermal process (iii) adiabatic process. (12)

OR

- III. (a) Explain first law of thermodynamics. Apply first law for an open system. (6)
(b) Find the difference between work done in compressing 0.28 m³ of air at a pressure of 1.4 bar to a volume of 0.028 m³, when the compression is adiabatic and polytropic with polytropic index $n=1.2$. (6)

- IV. Explain the working of 2 stroke SI engine. What are the advantages and disadvantages of a 2-stroke engine over a 4 stroke engine? (12)

OR

- V. Explain diesel cycle. Obtain the expression for air standard efficiency of a diesel cycle. (12)

- VI. (a) Distinguish between impulse and reaction turbine. (4)
(b) With the help of a neat sketch explain the working of Pelton turbine. (8)

OR

- VII. (a) Describe the working of a thermal power plant with suitable layout diagram. (8)
(b) What are the advantages and disadvantages of following power plants. (4)
(i) Hydroelectric (ii) Nuclear

- VIII. (a) Explain the process of gas welding using suitable diagrams. (6)
(b) What is meant by sand casting? Explain sand casting procedure. (6)

OR

- IX. (a) Derive an expression for length of belt for open belt drive. (8)
(b) Explain with diagrams (4)
(i) Extrusion (ii) Rolling
